

Funnel Analysis

I performed an exploratory analysis on the lead data and identified the following data quality issues:

- Some deals have a closing date (either won or lost) in the future → these were removed from the analysis.
- Some deals are “inactive” (i.e. only created, with no progression through the funnel) → due to the lack of additional information, these were retained in the analysis.
- Some deals have “disqualified” as the lost reason → these are likely not valid opportunities and were excluded from funnel metrics.

Additional assumptions:

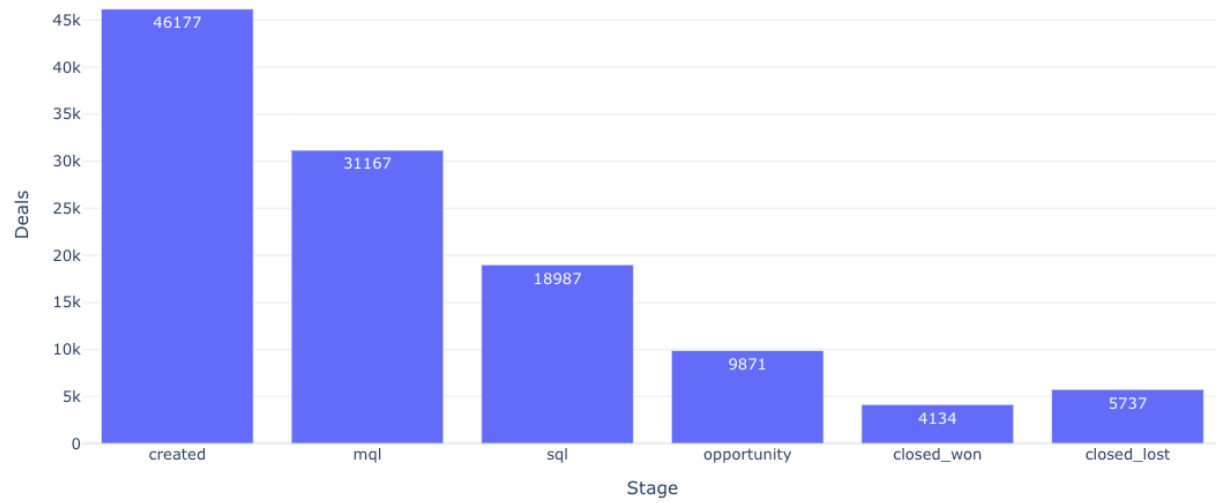
- Recently created deals may not have had sufficient time to progress through the funnel. As a result, they can bias conversion rates downward. However, I chose to retain them and account for this effect when interpreting the results.

Volumes and Metrics

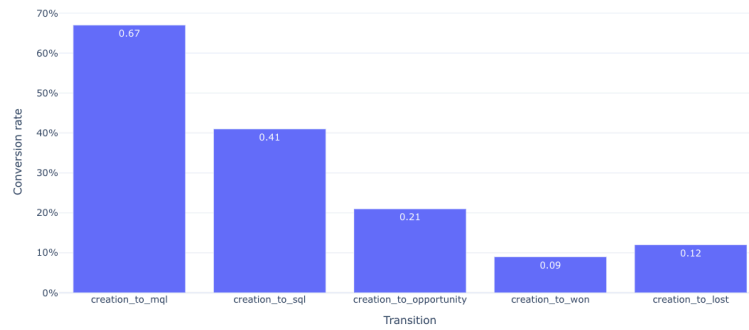
The following analysis presents:

- the volume of deals at each stage of the funnel
- step-by-step conversion rates
- end-to-end conversion rates
- conversion rate of the MQL and SQL phases
- average deal value by stage

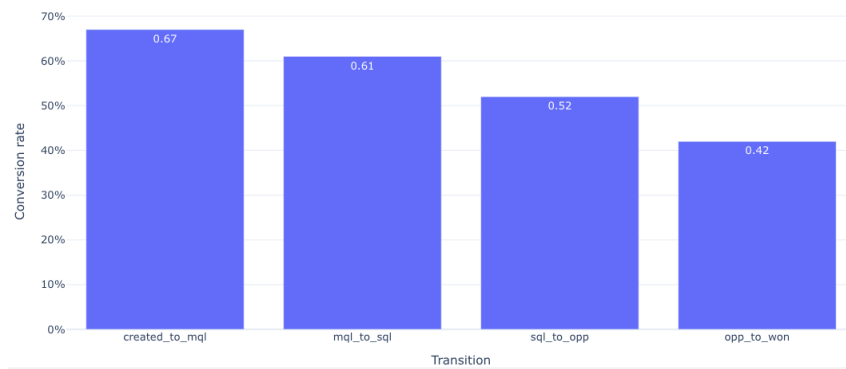
Funnel volumes



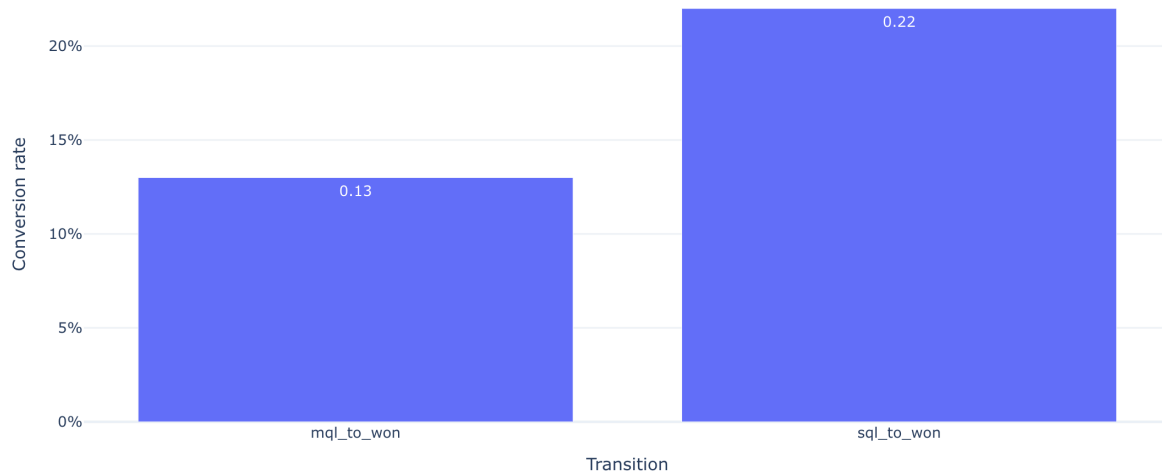
Created-based conversion rates



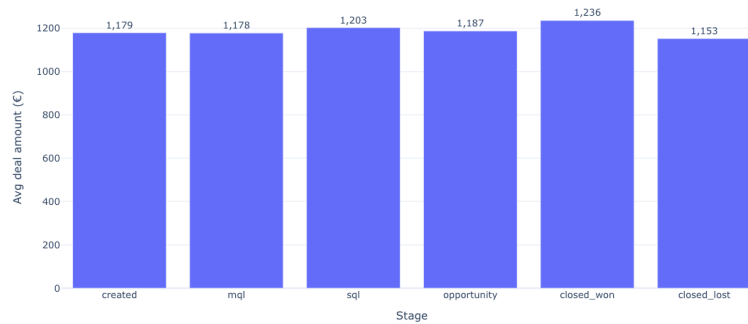
Step conversion rates



Stage-to-outcome conversion rates



Average deal amount by funnel stage



I also analysed the evolution of volumes and conversion rates over time:

- Deal volumes show a clear increasing trend over time.
- Conversion rates remain relatively stable across periods.

In terms of timing:

- The duration between stages typically ranges between 0 and 30 days.
- The final stage (Opportunity → Closed) is significantly longer, spanning up to 90 days.
- Overall, the full sales cycle can take up to ~160 days.

The results show:

- **Efficiency bottleneck – Opportunity → Closed Won:** this stage has the lowest conversion rate (~42%) and the longest duration, making it the hardest step in the funnel to successfully complete.
- **Volume bottleneck – SQL → Opportunity:** this is where the largest number of deals drop off (~9k deals), meaning improvements here would have the biggest impact on total conversions.
- **Early-stage qualification:** the large gap between MQL → Won (13%) and SQL → Won (22%) suggests that improving qualification quality earlier in the funnel could reduce downstream losses and improve overall performance.
- **Funnel robustness:** the average deal amount remains relatively stable (and even slightly higher for won deals), indicating that higher-value opportunities are not disproportionately lost. Additionally, despite increasing deal volumes over time, conversion rates remain stable, suggesting the funnel scales without degrading performance.

Focus on the Opportunity → Closed Phase

Since the final stage is both the slowest and the least efficient in terms of conversion, I conducted a deeper analysis of this step.

The most notable finding is that DEAL_DEMO_SCORE almost perfectly separates won and lost deals:

- Deals with a score ≥ 20 are consistently won
- Deals below this threshold are almost always lost

This indicates that, at this stage, the outcome is effectively already determined. Consequently, the marginal value of a predictive model is very limited, as a simple threshold on this variable would achieve near-perfect performance.

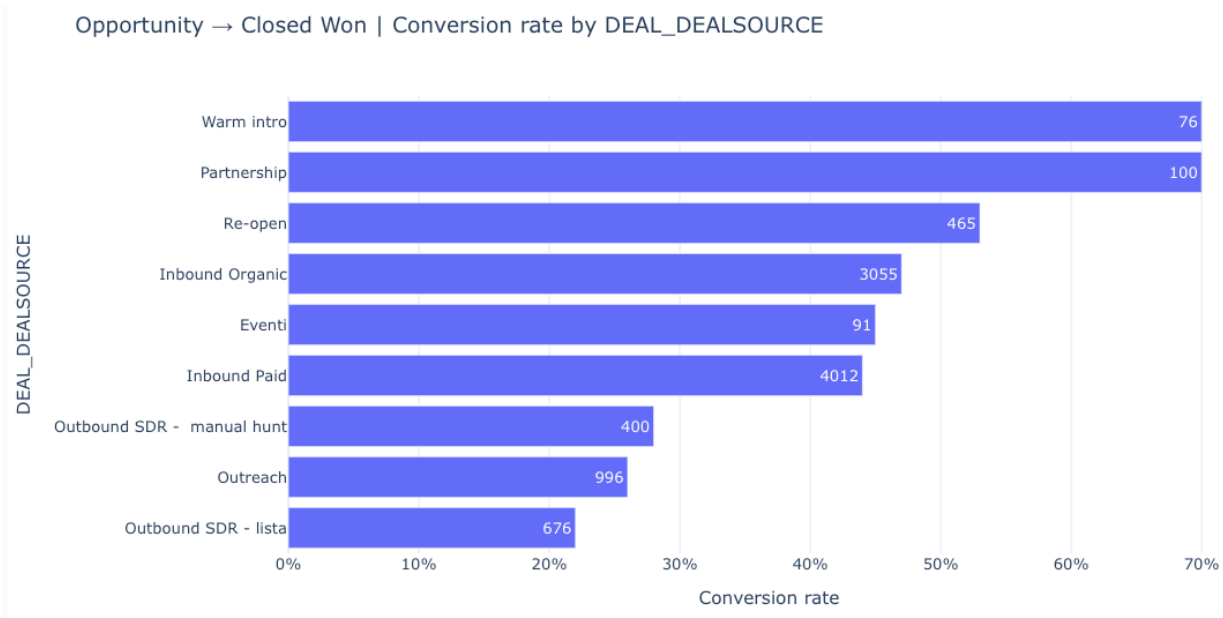
A critical consideration is **feature availability**:

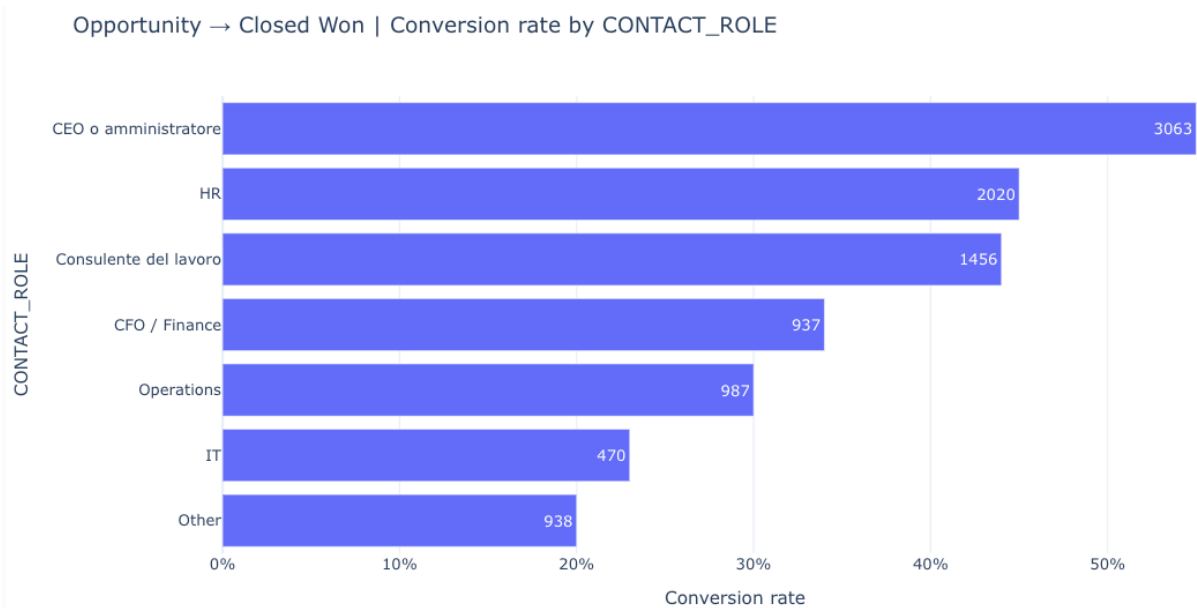
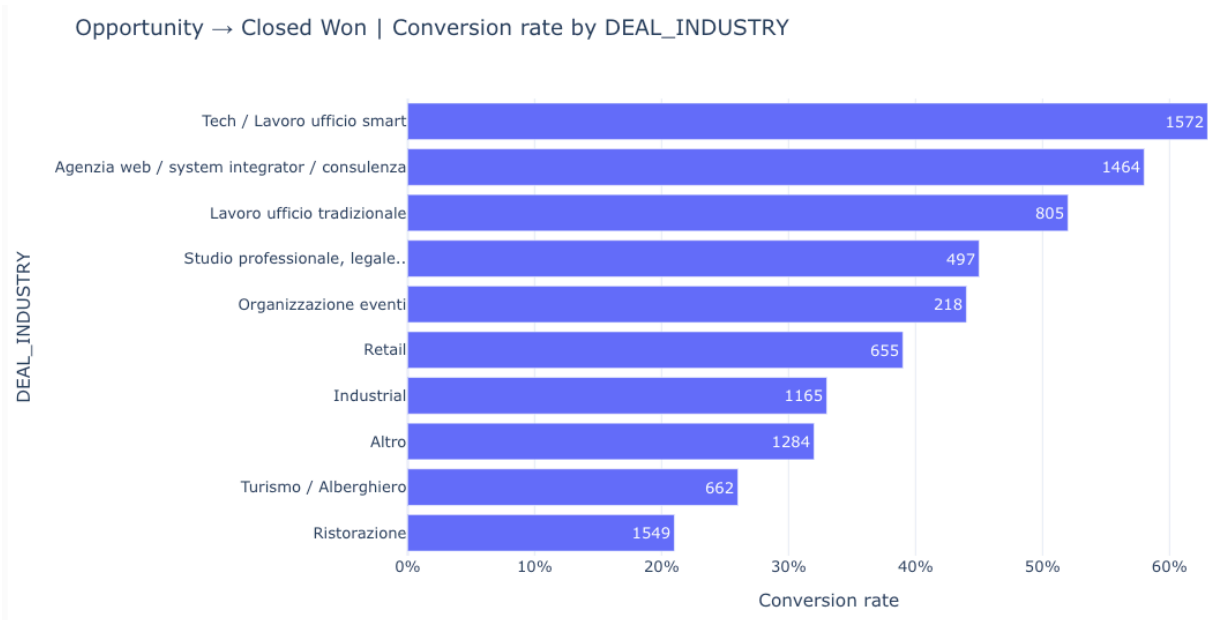
- If the demo score is computed using only information available at the Opportunity stage, it is a valid and highly predictive feature.
- If it relies on information revealed later in the process, it would introduce **target leakage** and should not be used for modelling before that point.

Even though prediction is trivial at this stage, analysing the drivers of conversion provides valuable business insights:

- Deal source: inbound channels outperform outbound ones
- Industry: some sectors (e.g. tech) convert significantly better than others (e.g. catering)
- Contact role: deals involving senior stakeholders (e.g. CEOs) show conversion rates above 50%
- Sales effort: deals contacted 5+ times exhibit higher conversion rates

The plots below show the distribution of conversion across these segments. The values inside each bar represent the number of deals in each category.





These variables are not necessary for prediction at this stage but provide useful business insights on what drives success.

Additionally, I analysed lost deal reasons:

- The most frequent reason is *"no answer"*
- These deals are contacted on average ~4 times
- Conversion rates increase significantly after 5+ contact attempts

This suggests that **increasing persistence on selected leads could improve overall performance.**

Focus on MQL → Closed and SQL → Closed Phases

I also analysed deals progressing from the MQL and SQL stages, where the largest drop-offs occur in absolute terms. In particular, I examined deal-level properties to understand which factors make a deal more likely to be successfully closed at these stages.

The same key drivers observed in the final stage remain relevant:

- deal source
- industry
- contact role

These variables consistently influence the likelihood of successfully closing a deal.

Insights

1. Increase persistence on high-potential leads

- Deals classified as *"no answer"* are contacted on average ~4 times, while conversion improves significantly after 5+ attempts.
- **Action:** introduce a minimum threshold of follow-ups (e.g. ≥5) before marking a deal as lost.

2. Prioritise high-converting segments

- Inbound leads, specific industries (e.g. tech), and senior roles (e.g. CEO) show higher conversion rates.
- **Action:** prioritise these segments through scoring or routing rules for the sales team.

3. Improve early-stage qualification (MQL → SQL)

- A large share of deals is lost in early stages (high absolute drop-off).

- **Action:** introduce lead scoring model to guide marketing and sales on resource allocation

4. Optimise the demo process

- The demo score perfectly determines the final outcome.
- **Action:** focus on improving demo effectiveness by standardising best-performing demo practices and monitor demo score over time.